

UEC720: ANTENNA AND WAVE PROPAGATION				
	L	T	P	Cr
	2	0	2	4.0
Course Objective: Students will be able to understand vector theory, antenna basic parameters, linear wire antennas, antenna arrays and their patterns, folded dipole, Yagi Uda, loop and Microstrip antenna, wave propagation over ground, through troposphere and ionosphere.				
Syllabus Introduction to Basic Antenna parameters: Radiation pattern, Radiation intensity, Beam width, Gain, Directivity, Polarization, Bandwidth, Efficiency, Side lobes, Side lobe level, Antenna Vector Effective Length and Equivalent Areas, Maximum Directivity and Maximum Effective Area, Friss Transmission Equation. Radiation Integrals and Auxiliary Potential Functions: Retarded vector and scalar potential, Vector Potential A for an Electric Current Source J, Vector Potential F for a Magnetic Current Source M, Electric and Magnetic Fields for Electric (J) and Magnetic (M) Current Sources. Linear Wire Antennas: Radiation from an infinitesimal small current element, Radiation from an elementary dipole (Hertzian dipole), Small Dipole, Finite length dipole, half wave dipole. Antenna Arrays: Two-Element Array, Broadside arrays, End fire arrays. N-Element Linear Array: Uniform Amplitude and Spacing, N-Element Linear Array: Directivity, N-Element Linear Array: Uniform Spacing, Non uniform Amplitude, Binomial Array, Chebyshev Arrays, Principle of pattern multiplication. Propagation of Radio Waves: Different modes of propagation: Ground waves, Space waves, Space wave propagation over flat and curved earth, Wave propagation in the Ionosphere, Critical frequency, Maximum usable frequency (MUF), Skip distance, Virtual height. Familiarization with Standards: <i>IEEE 145-2013, IEEE 1502-2007 and IEEE 211-1997</i> Laboratory Work: Drive antenna by voltage, Radiation pattern of half wave dipole, Radiation pattern of monopole, Effective height of antenna, Radiation pattern of capacitance and inductive loaded antenna, Directional radiation from two composite antennas, Radiation from conducting sheet with slot, Matching stub in antenna, Measure the SWR, Radiation polar diagram of directional antenna.				
Course Learning Objectives (CLO) The students will be able to: 1. Apply basic vector theory in solving antenna problems. 2. Examine various antenna parameters and apply those to the practical scenario. 3. Analyze wire antennas and asses their performance in real-world problems 4. Design and analyze antenna arrays for practical applications. 5. Identify and examine the characteristics of radio wave propagation for communication applications and effect on human health.				
Text Books 1. <i>Antenna Theory, Ballanis, John Wiley & Sons, 2003.</i> 2. <i>Antennas and Radio Propagation, Collins, R. E, McGraw-Hill, 1987</i>				

Reference Books

1. *Antennas, Kraus and Ronald Marhefka, John D., Tata McGraw-Hill, 2002.*
2. *Microwave & RF Design, Michael Steer, Sci.Tech Publishing, 2009.*